BECE MOCK EXAM
COMPUTING 2 & 1
Essay and Objective
2 hours

2 & 1

Name: .....

Index Number: .....

Cadidate's Signature .....

Date of Examination .....

Basic Education Certificate Mock Examination

COMPUTING 2 & 1 — 2 hours

Do **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions carefully. Write your* **name, index number, signature** *and* **date of examination** *in* **ink** *in the spaces provide above.*

This booklet consists of two papers. Answer Paper **2** *which comes first, in the booklet and Paper* **1** *on your Objective Test answer sheet. Paper* **2** *will last* **1** *hour* **15** *minutes. Do* **not** *start Paper* **1** *until you are told to do so. Paper* **1** *will last* **45** minutes. *At the end of the examination, submit the* **entire** *question paper to the invigilator.*

Any candidate who tears off any part of the question paper will be **severely** *penalized.*

Whether you answer all the questions in this paper **or** *not, hand in the* **entire** *question paper to the invigilator.*

For Examiner's Use Only	
Question Number	Mark

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PAPER 2 $1\frac{1}{4}$ hours

ESSAY

[60 marks]

This paper is in two sections: **A** *and* **B**. *Answer Question* **1** *in Section* **A** *and any other* **three** *questions in Section* **B**.

Answer **all** *the questions in the spaces provided in this question paper. At the end of the examination you should submit the* **entire** *question paper to the invigilator.*

Any candidate who tears off **any** *part of the question paper will be* **severely** *penalized.*

Whether you answer all the questions in this paper or not, hand in the **entire** *question paper to the invigilator.*

Credit will be given for clarity of expression and orderly presentation of material.

SECTION A

[24 marks]

Answer Question **1**

[*Compulsory*]

1. (a) A farmer wants to calculate the total cost of planting maize. He needs the cost per bag of fertilizer, number of bags, and labour cost.

(i) Identify **two** constants and **two** variables from the scenario above.

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[4 marks]

(ii) Suggest suitable names for **each** variable and constant for a program.

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[2 marks]

(b) The farmer buys 10 bags of fertilizer at GH¢ 50.00 **each** and hires labour for GH¢ 200.00.

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(i) Write the arithmetic expression to calculate the total cost using programming notation.

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[2 marks]

(ii) Show the same calculation in normal classroom Mathematics.

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[2 marks]

(c) Draw a flowchart using start/stop, process, and input/output symbols for these steps:

1. Clear the land
2. Plough the land
3. Apply fertilizer
4. Sow seed

[6 marks]

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(d) The farmer wants to decide whether to irrigate the farm:
- If it has rained → irrigation not needed
- If it has not rained → irrigation needed

Draw a flowchart with a decision symbol and label the "Yes" and "No" paths.

[8 marks]

SECTION B

[36 marks]

Answer **three** *questions* **only** *from this section.*

2. (a) A student compares a fourth generation computer with earlier generations of computers. State **two** advantages of fourth generation computers over first and second generation computers in everyday use.

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[4 marks]

(b) The processor is made up of different units that work together to execute instructions. Identify **two** main components of a processor and state the function of **each** component.

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[4 marks]

(c) How would you describe Mixed Reality in Artificial Intelligence to your friend?

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[2 marks]

(d) A supermarket notices that long queues form when cashiers manually type prices, but the queues reduce after barcode readers are introduced. In **two** ways, explain why the use of barcode readers improves the checkout process, making reference to the nature of the input device involved.

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[2 marks]

3. (a) An architectural firm produces large building designs, while a school displays lesson notes to a large class. Briefly explain why different output devices are required in these **two** situations, naming a suitable device for **each** case.

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[3 marks]

(b) A school ICT department wants to archive examination records so that they **cannot** be altered after submission. Identify the types of optical disc that would be **most** suitable for this purpose and state why.

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[3 marks]

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(c) With your knowledge in windows desktop, list any **four** icons on the charm bar.

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[4 marks]

(d) In an office, files are named randomly, causing confusion among staff members. State **two** reasons why file naming conventions are important for effective file management.

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[2 marks]

4. (a) A new computer user finds it difficult to locate open programs on the screen. In **four** ways, explain how the combined use of the desktop, taskbar, and taskbar buttons helps users organize and control their work.

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[4 marks]

(b) A family computer has separate accounts for parents and children. Explain **four** reasons why different user account levels are necessary on a shared computer system.

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[4 marks]

(c) A visually impaired student is admitted into a regular school that uses computers for learning. In **two** ways, explain how computer technology can be used to enable the student to participate fully in ICT lessons.

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[2 marks]

(d) A visually impaired teacher wants to read emails on her computer. Explain **two** technology she can use and how it helps.

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[2 marks]

5. (a) In a community, people rely on smart watches and mobile phones to monitor health and stay connected. In **two** ways, state how portable computing devices have influenced daily life in terms of communication and personal well being.

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[4 marks]

(b) Enumerate any **four** things that can be created with MS Publisher.

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[4 marks]

(c) Emmanuella could not differentiate between *template* and *flame* in MS Publisher. As a computing student, explain them to her.

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[4 marks]

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Answer **all** the *questions.*

Each question is followed by **four** *options lettered* **A** *to* **D**. *Find the* **correct** *option for* **each** *question and shade* in **pencil** *on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

Which of the following is an input device?

A. Monitor
B. Stylus
C. Projector
D. Speaker

The correct answer is Stylus, which is lettered B and therefore answer space B would be shaded.

[A] [B] [C] [D]

Think carefully before you shade the answer space. Erase completely any answers you wish to change.

Do all rough work on this question paper.

Now answer the following questions.

1. A school upgraded its computer lab with laptops that boot faster, handle multiple programs, and use less electricity. Which innovation is **mainly** responsible?

A. Magnetic drum
B. Microprocessor
C. Punch card
D. Vacuum tube

2. A pharmaceutical lab needs a computer to simulate molecular structures in seconds, a task that would take ordinary computers years. Which type of computer is suitable?

A. Desktop computer
B. Quantum computer
C. Tablet
D. Mainframe computer

3. Justice wants to resave his document under new name and different location. Which of these commands would **best** be used?

A. Save
B. Preview
C. File
D. Save AS

4. Internet Service Provider (ISP) provides access to the internet whiles International Standards Organizations develops

A. Open system interconnection network model.
B. Artificial Intelligence.
C. Protocol.
D. Robots.

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5. A group of webpages is a website but webpage is a/an
   A. electronic document.
   B. connection of computers.
   C. animation.
   D. graphic.

6. During a digital exam, a student uses a touchscreen while another uses a keyboard. The touchscreen **mainly** provides
   A. inability to run programs.
   B. direct interaction with icons and questions.
   C. more typing speed.
   D. slower input.

7. A supermarket replaces cashiers with barcode scanners at checkout. This improvement primarily
   A. replaces storage devices.
   B. makes keyboards better.
   C. slows down billing.
   D. increases speed and reduces errors.

8. In a clinic, patient files are scanned into the system for digital records. This is an example of
   A. output device.
   B. storage device.
   C. manual input.
   D. automatic input.

9. A physically challenged office worker uses voice recognition to input data. This software allows
   A. offline storage only.
   B. hands free data entry.
   C. automatic file deletion.
   D. replacing computers.

10. The brain or microprocessor of the robot processes information whiles gripper
   A. holds object.
   B. capture surrounding information.
   C. is for movement.
   D. convert electric energy to movement.

11. A teacher uses a small projector in a rural classroom to show educational videos. The **main** advantage is
   A. printing capability.
   B. large storage.
   C. portability and battery. operation.
   D. optical discs.

12. An architect needs to print a 3 meter floor plan. The **most** suitable device is
   A. Pico projector.
   B. Inkjet printer.
   C. Graphing plotter.
   D. Multimedia projector.

13. The application of Artificial Intelligence that finds solutions to problem by looking for patterns in database is known as
   A. Machine learning.
   B. Virtual reality.
   C. Augmented reality.
   D. Neural network.

14. Booting the computer is the act of turning on the computer whiles running an application is the act of
   A. restoring an application.
   B. opening an application.
   C. installing an application.
   D. uninstalling an application.

15. To demonstrate to a class the steps to be taken to add a footer in Microsoft word, click on

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A. home tab, then footer.
B. insert tab then footer.
C. page tayout then footer.
D. view tab, then footer.

16. With your knowledge in spreadsheet, which of the following is an example of Excel functions?
A. Argument
B. Count
C. Wrap text
D. Addition

17. Amma can't find her important document on her computer. As a computing student, it could be attributed to what attack?
A. Virus
B. Antivirus
C. Piracy
D. Worm

18. With your knowledge in spreadsheet, cell name is referenced by
A. active cell.
B. column and row.
C. column letter and row number.
D. row letter and column number.

19. Crawler moves through webpages to gather relevant information for a search result whiles database
A. runs the crawler.
B. request information.
C. stores information.
D. is an interface between user and the database.

20. A student switches between music, notes, and assignments without closing programs. Which windows feature allows this?
A. Start screen
B. Taskbar
C. Desktop
D. File explorer

21. A teacher previews multiple student documents without opening them fully. Which feature is being used?
A. Cloud backup
B. File extension
C. Temporal peeking
D. Tiles

22. Tiles on the Start screen help students
A. change file extensions.
B. store large files.
C. quickly access frequently used programs.
D. increase memory.

23. Thumbnails of open windows assist users by
A. reformatting drives.
B. deleting files automatically.
C. identifying contents of files before switching.
D. running cloud applications.

24. The desktop in Windows OS **mainly** serves as
A. cloud server.
B. printer interface.
C. storage device.
D. a working area for programs and files.

25. A student organises assignments in folders by subject. This practice is called
A. file deletion.
B. file organisation.
C. file extension.
D. file fragmentation.

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26. A shared computer has accounts for parents and children. Parents can install software while children can only view files. This illustrates
A. file extensions.
B. user account levels and permissions.
C. cloud storage.
D. input device usage.

27. Giving read only access to certain files ensures that
A. files are stored on smart cards.
B. files cannot be changed.
C. files are deleted automatically.
D. users cannot see files.

28. Using consistent file names such as "Math - Assignment1-2025" helps students
A. delete files faster.
B. organise and locate files easily.
C. slow computers.
D. increase storage.

29. A student opens a PDF document but **cannot** edit it. The file is **most** likely
A. backup disk.
B. cloud backup.
C. read only.
D. editable word document.

30. A visually impaired student uses software that reads text aloud. This software is
A. voice recognition.
B. word processor.
C. screen reader.
D. scanner.

31. Voice recognition software allows a physically challenged employee to
A. store files offline only.
B. delete files automatically.
C. replace computers.
D. enter data without using hands.

32. A student uses a smartwatch to receive notifications and monitor health. This illustrates
A. mainframe use.
B. portable computing devices improving daily life.
C. only desktop computing.
D. cloud storage.

33. During a power outage, a trader continues selling goods using a smartphone. This demonstrates
A. reliance on wired systems.
B. portability and battery independence.
C. limited storage.
D. need for manual input.

34. A student researches school assignments using a mobile phone from home. This **mainly** supports
A. distance and flexible learning.
B. manual record keeping.
C. offline communication.
D. only traditional learning.

35. Aso was very happy to see the outcome of what she entered into the computer on the monitor. As a computing student, the outcome on the monitor is referred to as
A. input.
B. processing.
C. displaying information.
D. processed data.

36. The content is one of the techniques for evaluating website which talks about the correctness of the information on the website. What about accuracy?

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A. How true the information on the website is
B. The person who wrote the website
C. How updated the website is
D. How the website works

37. As a computing student, which one of these is **not** a feature in MS Word?
A. Name box
B. Alignment
C. Line spacing
D. Format painter

38. The 3D printer is used to create any three-dimensional objects while holographic storage is used to
A. create animations.
B. store three-dimensional data.
C. print three-dimensional objects.
D. read electronic book.

39. File extension for MS Word is .docx, what about .pptx?
A. Spreadsheet
B. PowerPoint
C. Adobe acrobat
D. Plain text

40. A student put a code on his data to prevent his friend from having access to the data. In computing, this practice is termed as
A. debugging.
B. decoding.
C. decryption.
D. encryption.

END OF PAPER

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Answer all **the** *questions.*

Each question is followed by **four** *options lettered* **A** *to* **D**. *Find the* **correct** *option for each question and shade* **in pencil** *on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

Which of the following is **not** an element of design?

A. Dot
B. Harmony
C. Line
D. Texture

The correct answer is prevent them from rusting, *which is lettered B and therefore answer space C would be shaded.*

[A] [B] [C] [D]

Think carefully before you shade the answer space; erase completely any answers you wish to change.

Do all rough work on this question paper.

Now answer the following questions.

1. Akua looks at a butterfly and notices that the left wing perfectly matches the right wing. Which principle of design is she observing in nature?
   A. Contrast
   B. Proportion
   C. Balance
   D. Emphasis

2. During art class, Mr. Osei places a pot, a pineapple, and a knife on a table for students to look at and draw. This type of drawing is called
   A. landscape drawing.
   B. portrait drawing.
   C. still life drawing.
   D. imaginative drawing.

3. Using a yam cut in half, Kwame presses paint onto paper repeatedly. In printmaking and pattern making, the yam acts as his
   A. brayer.
   B. stamp.
   C. canvas.
   D. palette.

4. Abena takes a round ball of clay and uses her thumbs to press a hole in the center, squeezing the sides up to make a cup. This technique is known as
   A. coiling method.
   B. the pinch method.
   C. slabbing method.
   D. carving

5. Fati tears old newspapers, soaks them in water, and mixes them with starch to model a festival mask. The modelling material she created is
   A. terracotta.
   B. bronze.

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